

Google Cloud Run — Study Guide

Version 2 — high drift (major factual changes)

Autoscaling and Concurrency

Cloud Run now supports predictive autoscaling in addition to reactive scaling, using historical traffic patterns to pre-warm instances before demand spikes. Scale-to-zero is no longer the default; the new default minimum is 1 instance to eliminate cold starts for production workloads.

The default concurrency limit has been raised from 80 to 500 requests per instance. The maximum instance count default has increased from 100 to 1000 per service. Cold start latency has been reduced to under 200 ms for most container images via the new Rapid Startup feature, which pre-loads the container runtime.

Networking

Cloud Run now supports direct VPC egress without requiring a Serverless VPC Access Connector. Traffic to private IPs is automatically routed through the VPC, removing the connector cost and latency.

The new Private Service Connect integration replaces domain mapping for custom domains. Ingress now has a fourth option — "Internal and Cloud Armor" — which enables WAF rules for internal-facing services. IPv6 is supported end-to-end for new services.

Identity and Security

Cloud Run now integrates with Workload Identity Federation, replacing long-lived service account keys for cross-project authentication. Service accounts must be attached explicitly; the Compute Engine default service account can no longer be used with Cloud Run as of this release.

Binary Authorization is now enforced by default for all new Cloud Run services, requiring deployed images to be attested by a configured attestor. The Cloud Run Invoker role has been split into roles/run.invoker (HTTP) and roles/run.jobsInvoker (Jobs).

Pricing

Cloud Run pricing has moved to a new tier-based model:

- Tier 1 (0–1B requests/month): \$0.30 per million requests
- Tier 2 (1B+ requests/month): \$0.15 per million requests
- CPU: now charged per millisecond at \$0.000024 per vCPU-ms
- Memory: \$0.0000025 per GiB-ms

The flat monthly free tier has been replaced by a committed-use discount model. Minimum instances are now billed at 50% of the standard rate instead of full rate.